Nathan Gardner Hattersley

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Summary

Ph.D. candidate in Economics at the University of Texas at Austin with an M.S. in Statistics. Specialize in causal inference, pricing, demand estimation, and structural industrial organization using large-scale observational data. Experience spans antitrust litigation, policy research, machine learning, and production software engineering.

Education

- **University of Texas at Austin** — Austin, TX
  Ph.D. Economics (expected 2026); M.A. Economics — *Aug 2020–Present*
- **University of Arizona** — Tucson, AZ
  M.S. Statistics; B.A. Mathematics and Computer Science — *Aug 2013–May 2018*

Skills

- **Programming**: Python, R, Julia, SQL, TypeScript, Rust, Java, C, Stata, MATLAB, Bash
- **Economics & Statistics**: Causal inference, structural modeling, demand estimation, machine learning (XGBoost), nonparametrics, numerical optimization
- **Tools**: Pandas, tidyverse, Postgres/PostGIS, spatial analysis, Docker, Git, HPC, Make/Snakemake, LaTeX
- **Languages**: English (native); Spanish (proficient)

Professional Experience

- **Texas Office of the Attorney General** — Austin, TX
  Economics Fellow — *Jul 2024–Jun 2025*
  - Analyzed antitrust cases involving algorithmic collusion, online advertising auctions, and digital markets using industrial organization methods.
  - Drafted economic analyses and reviewed expert rebuttals supporting litigation strategy.
  - Identified a security vulnerability affecting a key litigation matter, informing defensive legal strategy.
  - Presented technical findings to attorneys and translated quantitative analyses into accessible legal arguments.
- **World Bank Group** — Remote
  Short-Term Consultant — *Feb 2021–Aug 2021*
  - Built statistical indices measuring political participation in Afghanistan using large-scale survey data.
  - Developed algorithms for ordinal survey aggregation and PCA-based index construction.
  - Authored technical documentation and presented methods to economists and policy researchers.
  - Developed and led internal training on R, tidy data, functional programming, and reproducible workflows.
- **JPMorgan Chase & Co.** — Houston, TX
  Software Engineer — *Jun 2018–Jun 2020*
  - Developed and maintained production trading systems in Python and TypeScript supporting interest rate derivatives.
  - Designed regulatory workflows and trade models for new financial instruments.
  - Migrated legacy platform to microservice architecture using React, GraphQL, and WebSockets.
  - Collaborated with traders, product managers, and engineers to deliver production software.

Selected Research

- **Dealer Consolidation and Vertical Relationships**: Causal analysis of automobile dealership consolidation using transaction-level data and geospatial analysis to estimate effects on prices and product availability.
- **Dynamic Pricing in Automotives**: Developed and estimated a dynamic structural model of manufacturer-dealer relationships. Combined structural estimation with machine learning (XGBoost) to recover underlying allocation and sales target policy of the manufacturer. Use estimates to recover welfare effects of dealer allocation policy.
- **Demand Elasticity and Curvature (Joint with Eugenio Miravete)**: Nonparametric estimation of consumer demand to study the relationship between elasticity, curvature, pass-through, and market power, with comparisons against widely used demand systems.

Additional Experience

- **Teaching**: Teaching assistant for Master's Econometrics, Industrial Organization, and Causal Inference. Led weekly review sessions, designed programming assignments, and taught statistical computing.
- **Research Computing**: Led workshops on high-performance computing for economics Ph.D. students and developed departmental guidance for parallel computing in Julia, Python, and MATLAB.